

From Linear RNNs to Nonlinear Random RNNs and Dynamical Mean-Field Theory (DMFT)

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These notes trace the development from linear recurrent neural networks (RNNs) to nonlinear random RNNs and the application of Dynamical Mean-Field Theory (DMFT). We discuss key assumptions and when DMFT is applicable.

Linear Recurrent Neural Network

We start with a linear RNN:

$$\tau \frac{du_i}{dt} = -u_i + \sum_j J_{ij} u_j + I_i^{\text{ext}},$$

or in vector form:

$$\tau \frac{d\vec{u}}{dt} = -\vec{u} + W\vec{u} + \vec{I}^{\text{ext}}.$$

If W is diagonalizable, write $W = V\Lambda V^{-1}$ with $\Lambda = \text{diag}(\lambda_0, \lambda_1, \dots)$. Let $\vec{v} = V^{-1}\vec{u}$ and $\vec{M}^{\text{ext}} = V^{-1}\vec{I}^{\text{ext}}$. Then the system decouples:

$$\tau \frac{d\vec{v}}{dt} = -\vec{v} + \Lambda\vec{v} + \vec{M}^{\text{ext}}(t).$$

For each mode:

$$v_i(t) = \frac{M_i^{\text{ext}}}{1 - \lambda_i} \left[1 - e^{-t/[\tau/(1-\lambda_i)]} \right] + \text{const.}$$

1 Random Asymmetric RNN with Nonlinearity

Now, consider a nonlinear RNN with random Gaussian couplings:

$$\tau \frac{du_i}{dt} = -u_i + \sum_{j=1}^N J_{ij} \phi(u_j),$$

with $\phi(u) = \tanh(u)$ and

$$\langle J_{ij} \rangle = 0, \quad \langle J_{ij} J_{kl} \rangle = \frac{g^2}{N} \delta_{ik} \delta_{jl}.$$

Key Concepts and Assumptions

Quenched Disorder

The couplings J_{ij} are drawn from a Gaussian distribution:

$$J_{ij} \sim \mathcal{N}\left(0, \frac{g^2}{N}\right),$$

but for a given system, these couplings are **frozen** (fixed) for that specific realization. This is called **quenched disorder**.

Self-Averaging Assumption

We assume that averaging over many units i in a given realization (for some ensemble) is similar to averaging over an ensemble of systems. This **self-averaging** property allows us to replace disorder averages with spatial averages within a single large system.

Mean-Field Calculation

We are interested in behaviors that are independent of any specific realization. The mean-field approach approximates interactions by their average effect, which becomes exact in the limit $N \rightarrow \infty$.

Effective Noise and Correlations

Define the effective input to unit i :

$$\eta_i(t) \equiv \sum_{j=1}^N J_{ij} \phi(u_j(t)).$$

We compute its correlation:

$$\begin{aligned} \langle \eta_i(t) \eta_i(t') \rangle &= \left\langle \sum_{j,k} J_{ij} J_{ik} \phi(u_j(t)) \phi(u_k(t')) \right\rangle \\ &= \sum_{j,k} \langle J_{ij} J_{ik} \rangle \langle \phi(u_j(t)) \phi(u_k(t')) \rangle \\ &= \sum_{j,k} \frac{g^2}{N} \delta_{jk} \langle \phi(u_j(t)) \phi(u_k(t')) \rangle \\ &= \frac{g^2}{N} \sum_j \langle \phi(u_j(t)) \phi(u_j(t')) \rangle. \end{aligned}$$

Define the population-averaged correlation:

$$r(t, t') \equiv \frac{1}{N} \sum_j \langle \phi(u_j(t)) \phi(u_j(t')) \rangle.$$

Then:

$$\langle \eta_i(t) \eta_i(t') \rangle = g^2 r(t, t').$$

Thus, each unit feels an effective noise with temporal structure given by $r(t, t')$.

Dynamical Mean-Field Theory (DMFT) Setup

For large N , the dynamics of a single unit can be approximated by:

$$\tau \frac{du}{dt} = -u + \eta(t),$$

with

$$\langle \eta(t) \eta(s) \rangle = g^2 \langle \phi(u(t)) \phi(u(s)) \rangle.$$

We assume stationarity: correlations depend only on $\tau = t - s$. Let $C(\tau) = \langle u(t) u(t - \tau) \rangle$ and $C_{\phi\phi}(\tau) = \langle \phi(u(t)) \phi(u(t - \tau)) \rangle$.

The equation for u implies:

$$\left(\frac{d}{dt} + 1 \right) u = \eta(t).$$

Act on the correlation function:

$$\left(\frac{d}{dt} + 1 \right) \left(\frac{d}{ds} + 1 \right) \langle u(t) u(s) \rangle = \langle \eta(t) \eta(s) \rangle.$$

Since C depends on $\tau = t - s$, $\frac{d}{ds} = -\frac{d}{d\tau}$. Thus:

$$\left(\frac{d}{d\tau} + 1 \right) \left(-\frac{d}{d\tau} + 1 \right) C(\tau) = \left(1 - \frac{d^2}{d\tau^2} \right) C(\tau) = g^2 C_{\phi\phi}(\tau).$$

So, the DMFT equation is:

$$\left(1 - \frac{d^2}{d\tau^2} \right) C(\tau) = g^2 C_{\phi\phi}(\tau).$$

Evaluating $C_{\phi\phi}(\tau)$ via Gaussian Integrals

Assume $u(t)$ is a Gaussian process with mean zero, variance $C_0 = \langle u^2 \rangle$, and covariance $C(\tau)$.

Write two Gaussian variables $x_1 = u(t)$, $x_2 = u(s)$ with

$$\langle x_1^2 \rangle = \langle x_2^2 \rangle = C_0, \quad \langle x_1 x_2 \rangle = C(\tau).$$

We can represent them as:

$$\begin{aligned} x_1 &= \sqrt{C_0 - \frac{C^2}{C_0}} z_1 + \frac{C}{\sqrt{C_0}} z_2, \\ x_2 &= \sqrt{C_0} z_2, \end{aligned}$$

where $z_1, z_2 \sim \mathcal{N}(0, 1)$ independent.

Then:

$$C_{\phi\phi}(\tau) = \iint \phi(x_1)\phi(x_2) Dz_1 Dz_2,$$

with $Dz = \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz$. This defines a function:

$$C_{\phi\phi}(\tau) = f_\phi(C_0, C(\tau)).$$

Potential Formulation

The DMFT equation can be written as:

$$\frac{d^2 C}{d\tau^2} = C - g^2 f_\phi(C_0, C).$$

Introduce a potential $V(C)$ such that:

$$\frac{d^2 C}{d\tau^2} = -\frac{\partial V}{\partial C}.$$

Then:

$$V(C) = -\frac{1}{2}C^2 + g^2 \Phi(C_0, C),$$

where

$$\frac{\partial \Phi}{\partial C} = f_\phi(C_0, C).$$

Homework: Price's Theorem

Show that for functions whose variables are Gaussian with a specified variance and covariance structure C_0 and C , we have

$$\frac{\partial \Phi}{\partial C} = f_{\Phi'}(C_0, C).$$

In other words,

$$\Phi = f_\Phi(C_0, C) = \iint \Phi(x_1)\Phi(x_2) Dz_1 Dz_2.$$

For $\phi(x) = \tanh(x)$,

$$\frac{d\Phi}{dx} = \phi(x) \quad \Rightarrow \quad \Phi(x) = \ln \cosh(x).$$

This also closes the DMFT equations. The system's steady state and dynamics are determined by solving for $C(\tau)$ self-consistently.

When Does DMFT Work? Key Conditions

DMFT is applicable under the following conditions:

1. **Large N limit:** The number of units must be large enough that statistical fluctuations become negligible.
2. **Quenched disorder:** The random couplings are fixed in time, allowing for self-averaging to occur.
3. **Self-averaging:** Properties of individual units converge to the ensemble average as $N \rightarrow \infty$.
4. **Mean-field interactions:** Each unit interacts with many others, making the effective input approximately Gaussian (by the Central Limit Theorem).
5. **Asymmetric couplings:** The J_{ij} are typically asymmetric, preventing symmetry-induced collective phenomena that would require different treatment.
6. **Stationarity:** We assume the system reaches a stationary state where correlations depend only on time differences.

Physical Interpretation

The DMFT approach effectively reduces an N -dimensional stochastic dynamical system to a **single stochastic differential equation** with self-consistent noise statistics. The key insight is that in the large N limit:

- The effective noise $\eta(t)$ becomes Gaussian
- Its correlation is determined by the population-averaged activity
- This creates a self-consistent loop that can be solved numerically

Limitations

DMFT may fail when:

- The system size is too small for self-averaging
- Strong correlations develop between units (e.g., synchronization)
- The network has special structure (symmetry, regular patterns)
- The dynamics are far from stationarity